d- $\sum_{n=1}^{\infty} \frac{3}{n(n+3)}$

$$\frac{a}{n} + \frac{b}{n+3} = \frac{a(n+3) + bn}{n(n+3)}$$

$$\rightarrow 3 = (a+b)n + a3$$

$$\Leftarrow \begin{cases} a+b = 0 \\ 3a = 3 \end{cases} \Rightarrow \begin{cases} b = -1 \\ a = 1 \end{cases}$$

$$\sum_{n=1}^{N} \frac{3}{n(n+3)} = \sum_{n=1}^{N} \frac{1}{n} + \frac{-1}{n+3}$$

$$= \sum_{n=1}^{N} \frac{1}{n} - \underbrace{\sum_{n=1}^{N} \frac{1}{n+3}}_{\sum_{n=3}^{N+3} \frac{1}{n}} = 1 + \frac{1}{2} + \frac{1}{3} + \sum_{4}^{N} 1/n - \sum_{4}^{N} 1/n - 1/N+1 - 1/N+2 - 1/N+3$$

$$= 1 + \frac{1}{2} + \frac{1}{3} - \underbrace{1/N+1 - 1/N+2 - 1/N+3}_{\xrightarrow[N\to+\infty]{} 0}$$

$$\rightarrow 1 + 1/2 + 1/3 = 6/6 \quad 3/6 + 2/6 = 11/6$$

$$\int \frac{3}{x(x+3)} = \int \frac{1}{x} - \int \frac{1}{x+3}$$

$$\frac{n}{(n+1)(n+2)(n+3)} = \frac{a}{n+1} + \frac{b}{n+2} + \frac{c}{n+3}$$

$$= \frac{a(n+2)(n+3) + b(n+1)(n+3) + c(n+1)(n+2)}{(n+1)(n+2)(n+3)}$$

$$\begin{aligned} n = {} & an^2 + a5n + 6a & = (a+b+c)n^2 \\ & + bn^2 + b4n + 3b & + (5a+4b+2c)n \\ & + cn^2 + c2n + 2c & + (6a+3b+2c) \end{aligned}$$

$$\begin{cases} a+b+c = 0 \\ 5a+4b+2c = 1 \\ 6a+3b+2c = 0 \end{cases}$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 5 & 4 & 2 \\ 6 & 3 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$n \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = n^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\frac{n}{(n+2)(n+3)} = a + \frac{b(n+1)}{(n+2)} + \frac{c(n+1)}{(n+3)}$$

$$a = \frac{-1}{1 \times 2} = -1/2$$

$$\frac{n}{(n+1)(n+3)} = b + \boxed{0}$$

$n = -2$

$$b = \frac{-2}{-1 \cdot 1} = +2$$

$n = -3$

$$\frac{n}{(n+2)(n+1)} = c + 0$$

$$c = \frac{-3}{(-1) \times (-2)} = \frac{-3}{2}$$

$$\sum_{n=1}^{\infty} \left[\frac{(-1/2)}{n+1} + \frac{2}{n+2} + \frac{-(3/2)}{n+3} \right]$$

$$= \frac{-1}{2} \sum_{n=1}^{\infty} \frac{1}{n+1} + 2 \sum_{1}^{\infty} \frac{1}{n+2} + \frac{-3}{2} \sum_{1}^{\infty} \frac{1}{n+3}$$

$$= -\frac{1}{2} \left(\frac{1}{2} + \frac{1}{3} + \sum_{n=1}^{\infty} \frac{1}{n+3} \right) + 2 \left(\frac{1}{2} + \sum_{1}^{\infty} \frac{1}{n+3} \right) + \frac{-3}{2} \sum_{1}^{\infty} \frac{1}{n+3}$$

$$= (1 \cdots 1) + \underbrace{\left(\frac{-1}{2} + 2 + \frac{-3}{2} \right)}_{= 0} \sum_{n=1}^{\infty} \frac{1}{n+3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n+2} = \sum_{n=1}^{\infty} \frac{1}{(n-1)+3} \quad \boxed{m = n-1} = \sum_{m=0}^{\infty} \frac{1}{m+3} = \sum_{n=0}^{\infty} 1/n+3$$

$$= -\frac{1}{4} - \frac{1}{6} + 1 = \frac{-6-4}{24} + 1 = \frac{-5}{12} + 1 = \frac{12-5}{12} = \boxed{= 7/12}$$

$$\sum_{1}^{\infty} a_n + \sum_{2}^{\infty} a_n - 2 \sum_{3}^{\infty} a_n$$

$$= a_1 + a_2 + \sum_{3}^{\infty} a_n + a_2 + \sum_{3}^{\infty} a_n - 2 \sum_{3}^{\infty} a_n$$

$$= a_1 + a_2 + a_2$$